CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 8: Remainder and Factor Theorems

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The factor theorem

1. Which factor is common to $x^2 - 4$ and $x^3 - x^2 - 4x + 4$? (a) $(x + 1)$ (b) $(x - 1)$ (c) $(x - 2)$ (d) $(x - 4)$

ICSE 2025, 1 mark

SOLUTION

$x^2 - 4 = (x - 2)(x + 2)$

$x^3 - x^2 - 4x + 4 = x^2(x - 1) - 4(x - 1) = (x - 1)(x - 2)(x + 2)$

Of the options, $(x - 2)$ divides both.

Answer: (c)

2 Factorising a cubic polynomial

2. $(x - 2)$ is a factor of $2x^3 - 7x^2 + kx - 2$. Find k, and then factorise the polynomial completely.

ICSE 2025, 3 marks

SOLUTION

By the factor theorem $f(2) = 0$: $16 - 28 + 2k - 2 = 0$, so $k = 7$.

Divide $2x^3 - 7x^2 + 7x - 2$ by $(x - 2)$: the quotient is $2x^2 - 3x + 1$.

$2x^2 - 3x + 1 = (2x - 1)(x - 1)$

Answer: $k = 7$; $2x^3 - 7x^2 + 7x - 2 = (x - 2)(2x - 1)(x - 1)$

3. $(x - 1)$ is a factor of $3x^3 + 8x^2 - 15x + k$. Find k, and then factorise the polynomial completely.

ICSE 2024, 3 marks

SOLUTION

$f(1) = 0$: $3 + 8 - 15 + k = 0$, so $k = 4$.

Divide $3x^3 + 8x^2 - 15x + 4$ by $(x - 1)$: the quotient is $3x^2 + 11x - 4$.

$3x^2 + 11x - 4 = 3x^2 + 12x - x - 4 = (3x - 1)(x + 4)$

Answer: $k = 4$; $(x - 1)(3x - 1)(x + 4)$

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4. Use the factor theorem to factorise $2x^3 - x^2 - 13x - 6$ completely. ICSE 2023, 4 marks

SOLUTION

Try $x = -2$: $f(-2) = -16 - 4 + 26 - 6 = 0$, so $(x + 2)$ is a factor.

Dividing by $(x + 2)$ gives the quotient $2x^2 - 5x - 3$.

$2x^2 - 5x - 3 = 2x^2 - 6x + x - 3 = (2x + 1)(x - 3)$

Answer: $(x + 2)(2x + 1)(x - 3)$

5. Use the factor theorem to factorise $6x^3 + 17x^2 + 4x - 12$ completely. ICSE 2020, 3 marks

SOLUTION

Try $x = -2$: $f(-2) = -48 + 68 - 8 - 12 = 0$, so $(x + 2)$ is a factor.

Dividing by $(x + 2)$ gives the quotient $6x^2 + 5x - 6$.

$6x^2 + 5x - 6 = 6x^2 + 9x - 4x - 6 = (2x + 3)(3x - 2)$

Answer: $(x + 2)(2x + 3)(3x - 2)$

6. Use the factor theorem to show that $(x - 2)$ is a factor of $x^3 + x^2 - 4x - 4$. Then factorise the polynomial completely. ICSE 2019, 3 marks

SOLUTION

$f(2) = 8 + 4 - 8 - 4 = 0$, so $(x - 2)$ is a factor.

Dividing by $(x - 2)$ gives the quotient $x^2 + 3x + 2 = (x + 1)(x + 2)$.

Answer: $(x - 2)(x + 1)(x + 2)$

7. Use the remainder theorem to factorise $2x^3 + 3x^2 - 9x - 10$. ICSE 2018, 3 marks

SOLUTION

Try $x = 2$: $f(2) = 16 + 12 - 18 - 10 = 0$, so $(x - 2)$ is a factor.

Dividing by $(x - 2)$ gives the quotient $2x^2 + 7x + 5 = (2x + 5)(x + 1)$.

Answer: $(x - 2)(x + 1)(2x + 5)$

3 The remainder theorem

8. What must be subtracted from $x^3 + x^2 - 2x + 1$ so that the result is exactly divisible by $(x - 3)$? (a) -31 (b) -30 (c) 30 (d) 31 ICSE 2024, 1 mark

SOLUTION

The number to subtract is the remainder on division by $(x - 3)$, which is $f(3)$.

$f(3) = 27 + 9 - 6 + 1 = 31$

Answer: (d)

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9. What must be added to $2x^3 - 3x^2 - 8x$ so that it leaves a remainder of 10 when divided by $2x + 1$? ICSE 2020, 3 marks

SOLUTION

Let k be added. The remainder on division by $2x + 1$ is the value at $x = -\frac{1}{2}$.

$$2\left(-\frac{1}{8}\right) - 3\left(\frac{1}{4}\right) - 8\left(-\frac{1}{2}\right) + k = -\frac{1}{4} - \frac{3}{4} + 4 + k = 3 + k$$

$3 + k = 10$

Answer: 7

10. Find the remainders when $x^3 + (kx + 8)x + k$ is divided by $x + 1$ and by $x - 2$. The two remainders add up to 1. Find k. ICSE 2019, 3 marks

SOLUTION

$f(x) = x^3 + kx^2 + 8x + k$

$f(-1) = -1 + k - 8 + k = 2k - 9$ and $f(2) = 8 + 4k + 16 + k = 5k + 24$

$(2k - 9) + (5k + 24) = 1$, so $7k = -14$

Answer: remainders $2k - 9$ and $5k + 24$; $k = -2$

11. What must be subtracted from $16x^3 - 8x^2 + 4x + 7$ so that the result has $2x + 1$ as a factor? ICSE 2017, 3 marks

SOLUTION

The number to subtract is the remainder on division by $2x + 1$, which is $f\left(-\frac{1}{2}\right)$.

$$f\left(-\frac{1}{2}\right) = 16\left(-\frac{1}{8}\right) - 8\left(\frac{1}{4}\right) + 4\left(-\frac{1}{2}\right) + 7 = -2 - 2 - 2 + 7 = 1$$

Answer: 1

4 Finding unknown constants

12. $(x - 2)$ is a factor of $x^3 - kx - 12$. Find k. (a) 3 (b) 2 (c) -2 (d) -3 ICSE 2023, 1 mark

SOLUTION

$f(2) = 0$: $8 - 2k - 12 = 0$, so $k = -2$.

Answer: (c)

13. Find a if $(x - a)$ is a factor of $3x^3 + x^2 - ax - 81$. ICSE 2023, 4 marks

SOLUTION

By the factor theorem $f(a) = 0$: $3a^3 + a^2 - a \times a - 81 = 0$

$3a^3 = 81$, so $a^3 = 27$.

Answer: $a = 3$

14. $(x+2)$ and $(x+3)$ are factors of x^3+ax+b. Find a and b. ICSE 2018, 3 marks

SOLUTION

$f(-2)=0$: $-8-2a+b=0$ ... (1)

$f(-3)=0$: $-27-3a+b=0$ ... (2)

Subtracting (2) from (1): $19+a=0$, so $a=-19$. From (1), $b=8+2a=-30$.

Answer: $a=-19$, $b=-30$

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